

1) Define cloud Computing and Explain definition.

Cloud computing is the on-demand delivery of Computing services - including servers, storage, database networking Software analytics and intelligence over the internet with pay-as-you-go pricing it allows users to access and uses its resources without owning or maintaining physical infrastructure

Four key characteristics

1) on demand self service

Users can procure Computing resources automatically without requiring human interaction with service providers.

2) Broad network Access

Resources are available over the network and accessed through standard mechanism (web browser, mobile app, APIs)

3) Resource pooling

Providers computing resources are pooled to serve multiple consumers using a multiple tenancy model

Q) Differentiate IaaS, PaaS, SaaS and XaaS

Real World Example:-

1) IaaS - AWS

- Ex:- launching a virtual server on AWS EC2
- user configuration operating system security application
- provider manages physical hardware

2) PaaS - Google App engine

- Ex:- deploying a python/java
- user configuration : Application code only
- provider manages : OS, runtime

3) SaaS - Salesforce CRM

- Example : using Salesforce for customer
- user configures : settings
- provides : Everything

4) XaaS

Example : Flight Reservation

Service : custom application

provider manages Everything while user focuses on building.

3) Role of Server Virtualization in cloud computing

Server Virtualization is a technology in cloud computing that divides a single physical server into multiple isolated virtual machines (VMs) each running its own operating system and application.

physical server:

1- Hypervision (Vmware, Xen, Kvm)

VM₁ (windows, App A)

VM₂ (Linux, App B)

VM₃ (Linux, App C)

VM₄ (windows App C)

Virtualization

vs Cloud Computing

Aspect

virtualization

cloud computing

Def

Technology that created
out hardware

service model

Focus

Technical abstraction

Business model

Scope

Single physical layer.

Else IT

management

Hypervision and VM
management

Self service

u) Role of data centers in delivering cloud services
A data center is a physical facility that houses computers, system storage system, networking equipment and related infrastructure for storing, processing and management.

Component	Function
Servers	Computing power
Storage	Data storage
Networking	Switches, routers, load
Power system	UPS, generators, power distribution
Cooling	Air conditioning system
Security	Physical and cyber security measures

Roles in cloud services

- 1) Physical Foundation
- 2) Redundancy and Reliability
- 3) Global Reach
- 4) Economics of Scale
- 5) Security.

2) Evolution of cloud computing

Phase 1:- Hardware

Evolution (1940-1990s)

Era	Development	Impact
1940-1950	mainframe computers	large, expensive machine batch processing
1960's	Time sharing systems	multiple user shared central computers
1970-1980	personal	shifted computing power
1990	client server	web services

Phase 2 & 3

1995	web browsing	universal access
1999	Virtual Scaled	server virtual introduced
2002	ACOS	Began offered Storage on Compute
2006	Fc2 Fc3	Load / pass
2008	Google App	paas
2011	IBM	Enterprise
2015	AI, Edge	next generation